

Project Report

Transcriptomic Analysis of Breast Cancer - Differential Expression, Pathway Enrichment, PPI Network, and Candidate Biomarker Discovery in GSE42568

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Dataset Description:

Parameters	Details
Accession Number	GSE42568
Cancer Type	Breast Cancer
Organism	Homo sapiens
Tumor Samples	104 breast cancer biopsies
Normal Samples	17 normal breast tissues
Data Source	NCBI Gene Expression Omnibus
Data Type	Microarray gene expression (log2 - normalized)

Project Question:

Which genes are significantly differentially expressed between breast cancer tumor tissue and normal breast tissue in GSE42568, and which candidate genes show potential as diagnostic biomarkers based on differential expression strength, pathway relevance, protein interaction network centrality, and classification performance?

Methods:

1. Data Preparation

The GSE42568 series matrix file was downloaded from NCBI GEO and parsing was done using Python in Google Colab. The information about the expression matrix was taken from the file; 54,675 probe rows and 121 sample columns. Sample metadata was obtained from the header section to obtain source labels for each sample.

Samples were labeled as Tumor (104 samples: Breast tissue, cancer) or Normal (17 samples: Breast tissue, normal). The metadata index was aligned with the expression matrix column order. No missing values were found in the expression matrix and data was already in log2-normalized form.

2. Differential Expression Analysis

For 54,675 probes present in the dataset, a Welch two-sample t-test was used to compare tumor versus normal expression values individually. Probes with zero variance in either group (1,829 probes) were excluded from testing. A total of 52,846 probes were tested.

Log2 fold change was calculated by subtracting the mean normal expression from the mean tumor expression for each probe. The Benjamini-Hochberg false discovery rate (FDR) correction was applied to all valid p-values to control for multiple testing across 52,846 probes.

Significant differentially expressed genes (DEGs) were defined using the thresholds: $FDR < 0.05$ AND $|\log_2 \text{fold change}| > 1$, where a log2FC threshold of 1 corresponds to a minimum 2-fold expression change.

3. Visualization

Three visualization images were produced. A volcano plot displaying log2FC versus $-\log_{10}(FDR)$ for all probes, a clustered heatmap of the top 50 DEGs by FDR, with hierarchical clustering of both probes and samples, a PCA plot using all probes to assess global separation between tumor and normal samples.

4. Functional Enrichment Analysis

Significant DEG gene symbols were obtained by querying the MyGene.info REST API using Affymetrix probe IDs as reporter identifiers. Upregulated and downregulated gene lists were submitted separately to the Enrichr API for GO Biological Process 2023 and KEGG 2021 Human pathway enrichment analysis. Results were filtered and visualized as horizontal bar plots showing $-\log_{10}(p\text{-value})$.

5. PPI Network Analysis

The top 50 upregulated and top 50 downregulated DEGs by FDR were submitted to the STRING API to retrieve protein-protein interactions. The interaction network was built using NetworkX and visualized with node sizes proportional to interaction degree. Hub genes were identified as genes with the highest interaction degree.

6. Candidate Biomarker Selection and Validation

Eight candidate biomarkers were selected based on differential expression strength; enrichment in cancer-relevant pathways; hub gene status in the PPI network; literature survey in breast cancer biology. A Random Forest classifier was trained on the 8-gene expression panel using a 75/25 stratified train-test split. Model performance was evaluated using ROC-AUC, confusion matrix, sensitivity, specificity, and 5-fold stratified cross-validation.

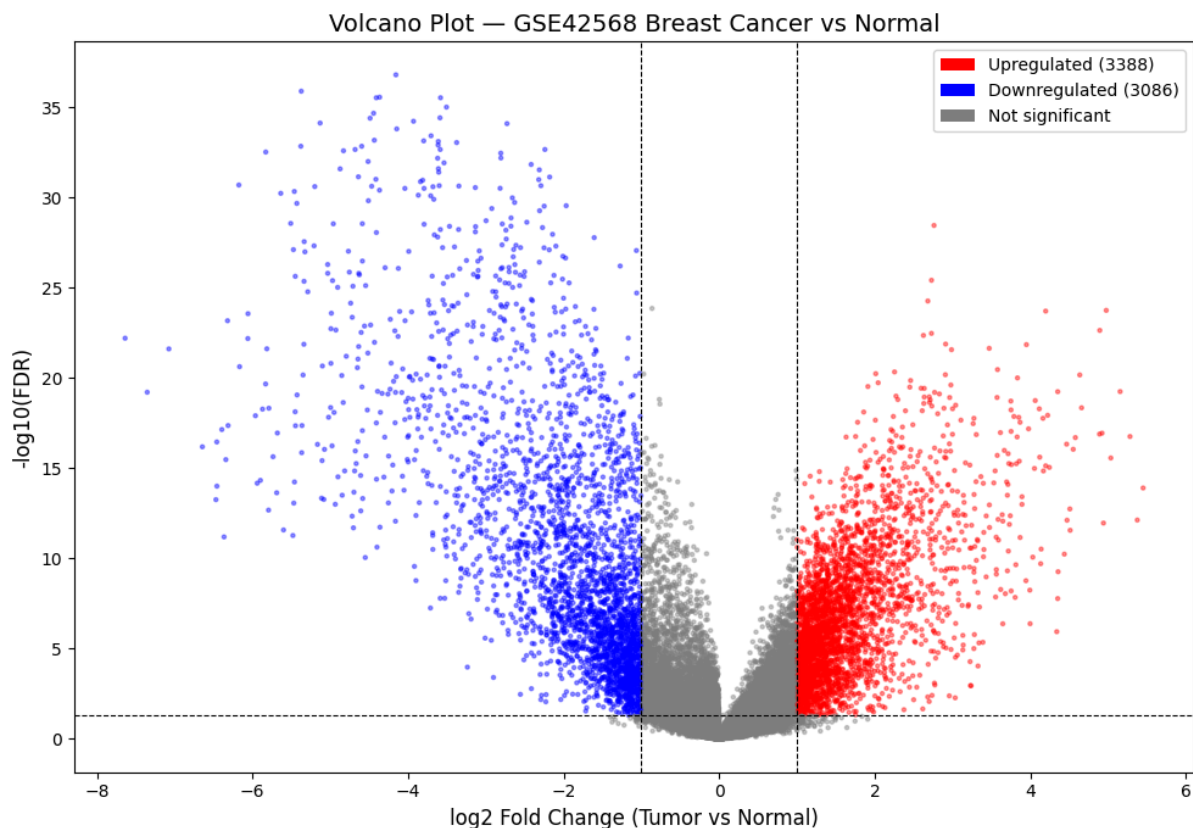
Result:

1. Differential Expression Analysis

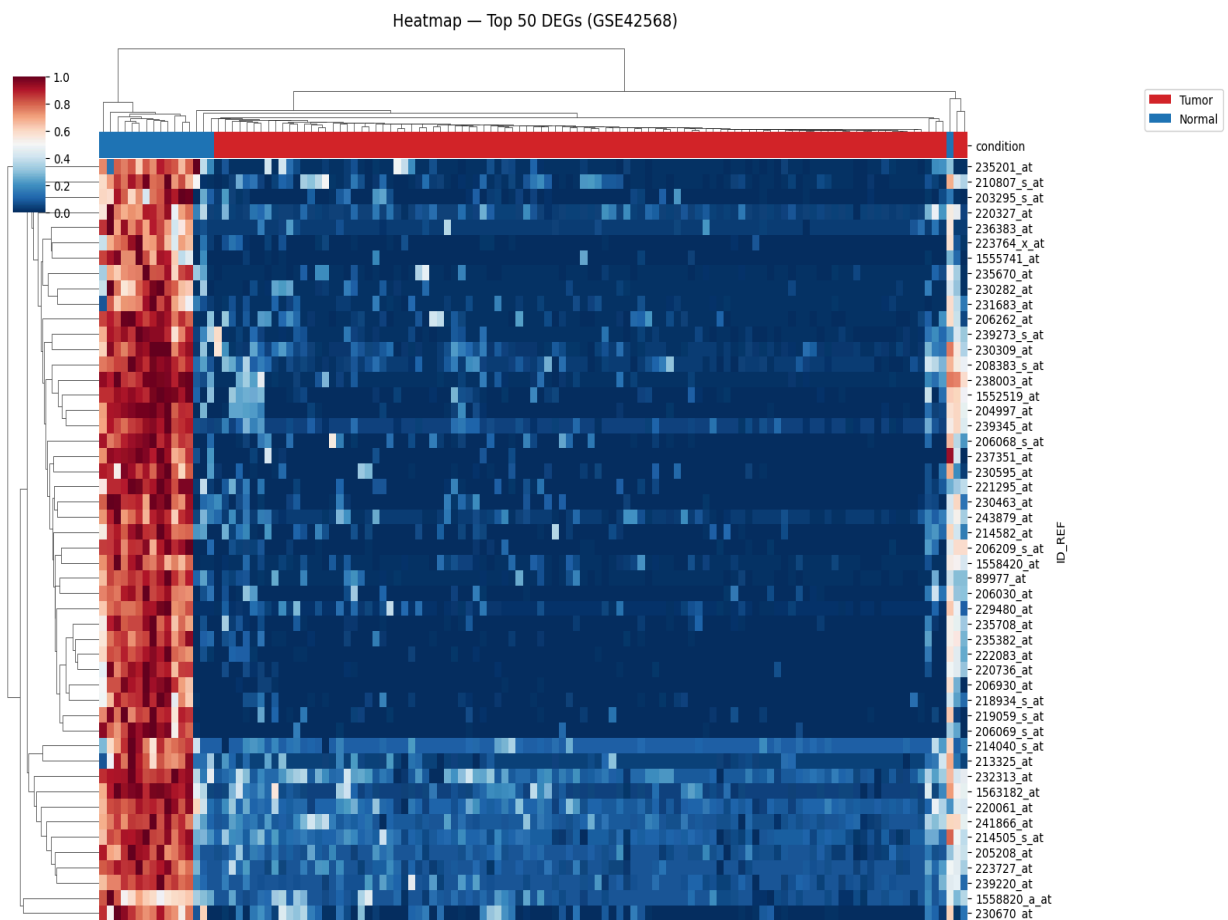
After excluding 1,829 zero-variance probes, 52,846 probes were tested. Applying thresholds of $FDR < 0.05$ and $|\log_2FC| > 1$ identified 6,474 significant DEGs: 3,388 upregulated and 3,086 downregulated in tumor compared to normal breast tissue. The top upregulated genes included KRT19, EPCAM, COL11A1, and CDH1, while the most strongly downregulated genes included FABP4 ($\log_2FC = -7.64$, $FDR = 6.26e-23$), LEP ($\log_2FC = -7.35$), and RBP4 ($\log_2FC = -7.08$), reflecting the replacement of normal adipose breast tissue by tumor cells.

2. Visualization

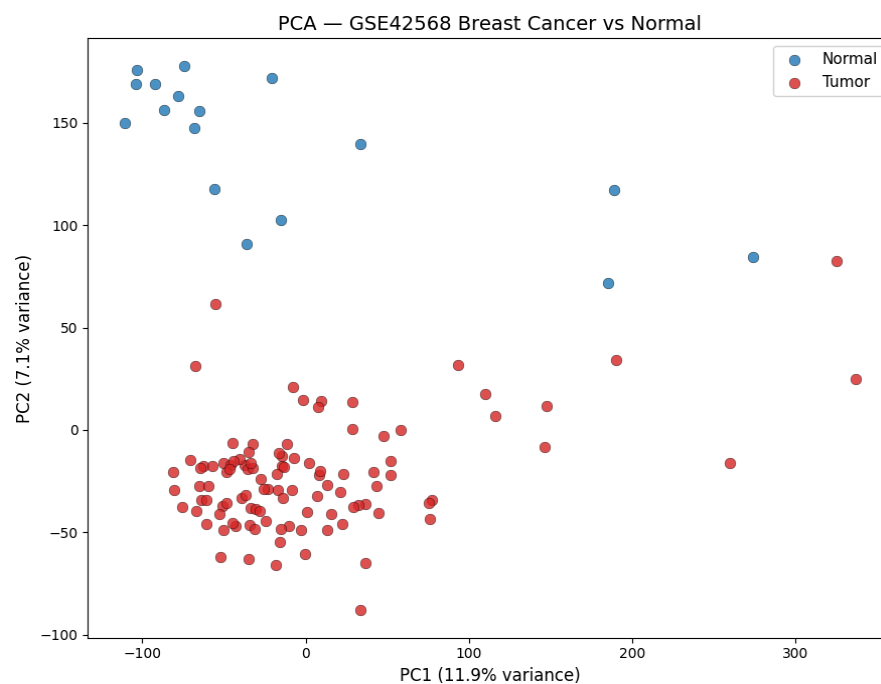
Volcano Plot: The volcano plot showed a symmetric distribution of upregulated and downregulated probes. Several probes reached $-\log_{10}(FDR)$ values above 30. The strongest downregulated signals had larger fold changes than the strongest upregulated signals.



Heatmap: Hierarchical clustering of the top 50 DEGs



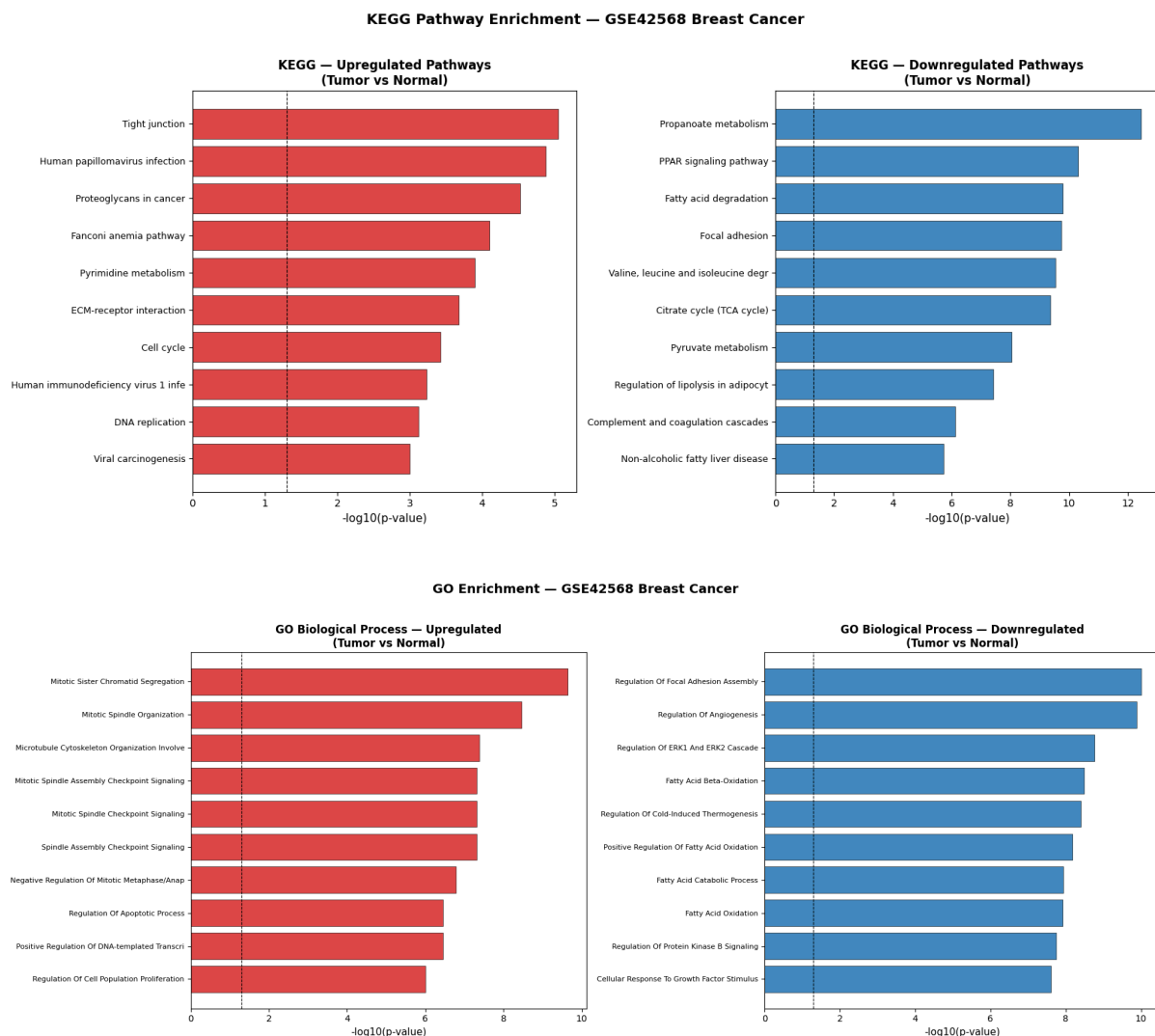
PCA Plot: PC1 (11.9% variance) and PC2 (7.1% variance) showed visible separation between normal (blue) and tumor (red) samples.



3. Functional Enrichment and Pathway Analysis

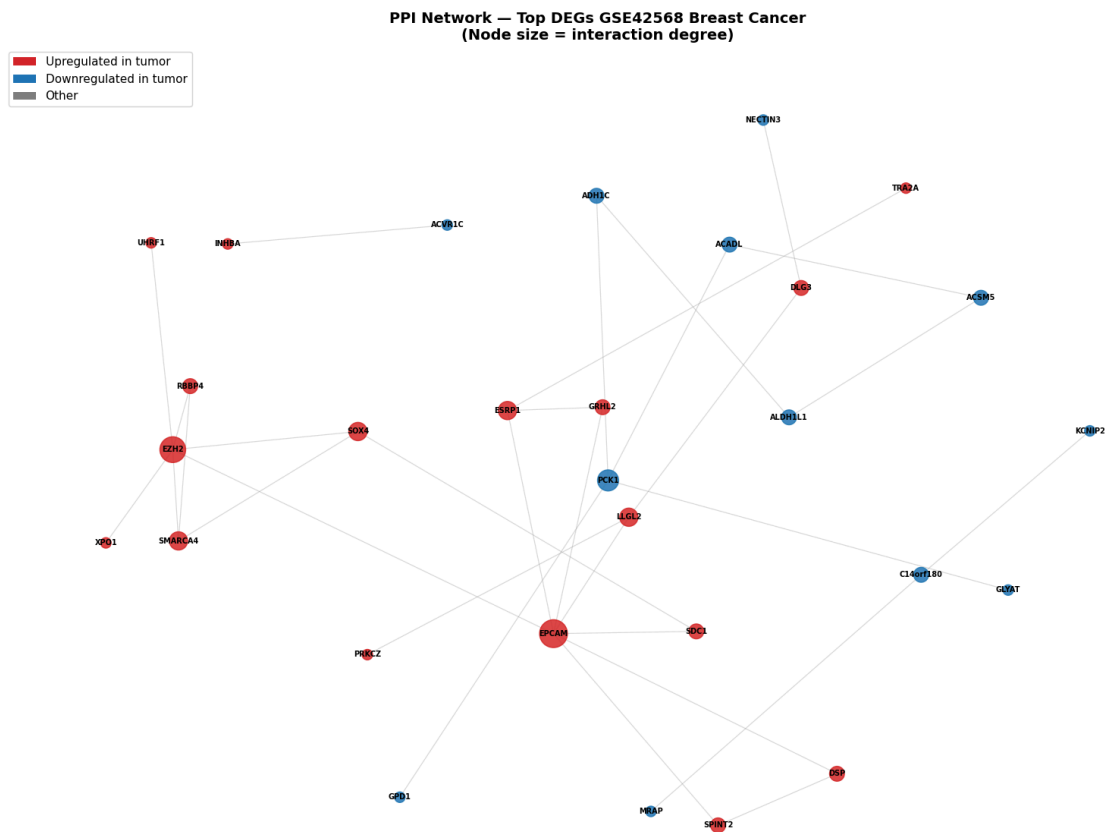
Upregulated pathways: GO Biological Process analysis revealed strong enrichment in Mitotic Sister Chromatid Segregation, Mitotic Spindle Organization, and Spindle Assembly Checkpoint Signaling. KEGG analysis identified enrichment in Tight Junction, Proteoglycans in Cancer, Fanconi Anemia Pathway, ECM-receptor interaction, and Cell Cycle.

Downregulated pathways: Downregulated genes were most strongly enriched in Propanoate Metabolism, PPAR Signaling, Fatty Acid Degradation, and TCA Cycle, reflecting the loss of normal adipose tissue metabolism as tumor cells replace normal breast cells.



4. PPI Network Analysis and Hub Genes

The STRING PPI network identified two major hub genes: EPCAM (degree 7) and EZH2 (degree 6).



Biological Interpretation and Clinical Insights:

1. Uncontrolled Proliferation

The most upregulated GO terms are all related to cell division: mitotic spindle organization, chromatid segregation, and spindle assembly checkpoint signaling. Upregulated genes including TOP2A, MKI67, and MELK are all established proliferation markers. TOP2A is a target of anthracycline chemotherapy and MKI67 is routinely used in clinical pathology as the Ki-67 proliferation index. The upregulation of these genes confirms that tumor cells are dividing at an elevated rate compared to normal breast tissue.

2. Loss of Epithelial Identity and EMT

Upregulated genes including SOX4, ESRP1, EZH2, and EPCAM are associated with epithelial plasticity and epithelial-mesenchymal transition (EMT). EZH2 is a polycomb repressor that silences tumor suppressor genes through histone methylation and is overexpressed in aggressive breast cancer subtypes. SOX4 promotes EMT and metastatic behavior. The downregulation of GATA3, a transcription factor that maintains luminal breast epithelial, indicates a difference from normal epithelial differentiation toward a more aggressive phenotype.

3. Replacement of Normal Adipose Tissue

The dominant downregulated signal reflects the biology of breast tissue composition. Normal breast tissue contains a substantial adipose component, and genes such as FABP4, LEP, PLIN1, RBP4, LPL, and CD36 are markers of adipocytes and lipid metabolism. Their downregulation shows the physical replacement of fat-containing normal breast stroma by tumor cell mass, supported by strong enrichment in PPAR signaling, fatty acid degradation, and lipolysis regulation pathways.

4. Genomic Instability and DNA Damage Response

The enrichment of the Fanconi Anemia pathway and DNA replication in upregulated KEGG terms indicates elevated DNA damage response activity in tumor samples, consistent with the genomic instability that characterizes breast cancer. The Fanconi Anemia pathway is involved in DNA interstrand crosslink repair, and its dysregulation is associated with cancer predisposition and chemotherapy sensitivity.

Candidate Biomarker Summary and Validation Evidence:

Gene	Description	Literature survey
EPCAM	Top PPI hub (degree 7) high log ₂ FC and FDR	EpCAM overexpression is well-validated in breast cancer; used clinically for circulating tumor cell detection
EZH2	Second PPI hub (degree 6); epigenetic oncogene, consistently upregulated	EZH2 overexpression associated with poor prognosis in breast cancer; target of tazemetostat therapy
TOP2A	Strong upregulation; cell cycle and DNA replication pathways	TOP2A amplification linked to HER2+ breast cancer; anthracycline chemotherapy sensitivity marker
MKI67	Standard clinical proliferation marker; upregulated in tumor	Ki-67 (MKI67) is the standard clinical proliferation index used in breast cancer pathology grading
SOX4	PPI hub (degree 3); EMT driver	SOX4 promotes breast cancer metastasis and poor prognosis; validated in multiple independent studies

CDH1	Focal adhesion pathway; loss of E-cadherin is hallmark of breast EMT and lobular carcinoma	CDH1 loss is the defining molecular feature of invasive lobular breast carcinoma
GATA3	Luminal breast identity marker; strong downregulation; loss in aggressive subtypes	GATA3 loss associated with poor prognosis; marker of luminal A/B subtype in PAM50 classification
FABP4	Strongest downregulated gene ($\log_2FC = -7.64$); PPAR signaling; adipose tissue loss	FABP4 loss reflects adipose-to-tumor tissue transition in breast microenvironment

References and Dataset Links

Dataset:

GSE42568: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE42568>

NCBI GEO: <https://www.ncbi.nlm.nih.gov/geo/>

GPL570 Platform: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GPL570>

Tools and Databases:

Enrichr: <https://maayanlab.cloud/Enrichr/>

MyGene.info: <https://mygene.info/>

STRING: <https://string-db.org/>

Human Protein Atlas: <https://www.proteinatlas.org/>

cBioPortal: <https://www.cbioportal.org/>

Key Literature:

Subramanian A, et al. Gene set enrichment analysis. PNAS, 2005.

Varambally S, et al. EZH2 in cancer progression. Nature, 2002.

Goldhirsch A, et al. Personalizing breast cancer treatment. Annals of Oncology, 2013.

Benjamini Y, Hochberg Y. Controlling the false discovery rate. Journal of the Royal Statistical Society, 1995.